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Unsymmetric Primal dual Problem.

$$Ax = b, x \geq 0.$$

↳ unsymmetric form.

The dual of the unsymmetric problem will have variables in unrestricted in sign.

$$\begin{cases} \text{Max } Z_x = cx \\ \text{Subject } Ax = b, x \geq 0 \end{cases} \rightarrow \text{dual problem}$$

$$\text{Min } Z_w = b'w$$

$$\text{Subject } A'w \geq c'$$

where w are unrestricted variables in sign.

Mixed type problem

$$\text{Max } Z = 3x_1 + 2x_2$$

$$\text{Subject } \left. \begin{aligned} 3x_1 + 2x_2 &\leq 2 \\ 2x_1 + 3x_2 &\leq 7 \end{aligned} \right\}$$

$$\boxed{5x_1 - 2x_2 = 5} \rightarrow$$

Any unsymmetric or mixed type problem
can be transform into standard.

$$\underline{5x_1 - 2x_2 = 5}$$

$$\left. \begin{array}{l} 5x_1 - 2x_2 \leq 5 \\ \text{and } 5x_1 - 2x_2 \geq 5 \end{array} \right\}$$

$$\Rightarrow \left. \begin{array}{l} 5x_1 - 2x_2 \leq 5 \\ -5x_1 + 2x_2 \leq -5 \end{array} \right\}$$

$$\text{Constraints: } \left. \begin{array}{l} 3x_1 + 2x_2 \leq 2 \\ 2x_1 + 3x_2 \leq 7 \\ 5x_1 - 2x_2 \leq 5 \\ -5x_1 + 2x_2 \leq -5 \end{array} \right\}$$

Thm: If k -th constraint of a primal
be an eqn, the k -th variable of the
corresponding dual will be unrestricted in
sign.

primal

$$\text{Max } Z_x = 2x_1 + 3x_2 - x_3$$

$$\text{Subject to } x_1 + 4x_2 + 2x_3 \leq 7$$

(2)

$$x_1 - 3x_2 + x_3 = 4$$

$$x_1, x_2, x_3 \geq 0$$

Mixed type

$$x_1 - 3x_2 + x_3 \leq 4$$

$$-x_1 + 3x_2 - x_3 \leq -4$$

constraint:

$$x_1 + 4x_2 + 2x_3 \leq 7$$

$$x_1 - 3x_2 + x_3 \leq 4$$

$$-x_1 + 3x_2 - x_3 \leq -4$$

$$A = \begin{bmatrix} 1 & 4 & 2 \\ 1 & -3 & 1 \\ -1 & 3 & -1 \end{bmatrix} \quad b = \begin{bmatrix} 7 \\ 4 \\ -4 \end{bmatrix}$$

$$c = (2, 3, -1)$$

Dual problem: $\text{Min } Z_w = b'w$

$$= 7w_1 + 4w_2' - 4w_2''$$

$w_1, w_2', w_2'' \geq 0$ be variables for dual problems.

$$A'w \geq c'$$

$$w_1 + w_2' - w_2'' \geq 2$$

$$4w_1 - 3w_2' + 3w_2'' \geq 3$$

$$2w_1 + w_2' - w_2'' \geq -1$$

Dual problem

$$\left. \begin{aligned} w_1 + (w_2' - w_2'') &\geq 2 \\ 4w_1 - 3(w_2' - w_2'') &\geq 3 \\ 2w_1 + (w_2' - w_2'') &\geq -1 \end{aligned} \right\} \begin{aligned} w_2' - w_2'' \\ = w_2 \end{aligned}$$

$$\left. \begin{aligned} w_1 + w_2 &\geq 2 \\ 4w_1 - 3w_2 &\geq 3 \\ 2w_1 + w_2 &\geq -1 \end{aligned} \right\}$$

$$w_2 = w_2' - w_2'', \quad w_2', w_2'' \geq 0$$

$$\left. \begin{aligned} \text{If } w_2' > w_2'' \text{ then } w_2 > 0 \\ \text{If } w_2' < w_2'' \text{ then } w_2 < 0 \end{aligned} \right\}$$

$\Rightarrow w_2$ is unrestricted in sign.

Then: If any variable of the primal problem is unrestricted in sign then the corresponding dual constraint will be strictly an eqn.

Revised Simplex Method